2D Linear Advection

PROBLEM SETUP

Equations

$$\frac{\partial u}{\partial t} + \mathbf{a} \cdot \nabla u = 0$$

$$\mathbf{a} = (a_y, a_z)$$

$$\Omega = [0,1] \times [0,1]$$

$$u(t,0,z) = u(t,1,z)$$

$$w(t,y,0) = w(t,y,1) = 0$$

Explanations

- Governing equation: **2D linear advection**
- Scalar field **u** transported by velocity **a**
- Computational domain: **unit square**
- Advection field **a = (a_y, a_z)**

- **y direction:** periodic boundary
- **z direction:** wall-bounded
- No penetration condition at walls

NUMERICAL STRATEGY

Equations

Operator Splitting

$$u^* = u^n - a_y \frac{\partial u}{\partial y} \Delta t$$

$$u^{n+1} = u^* - a_z \frac{\partial u}{\partial z} \Delta t$$

Upwind Discretization

If $a > 0$

$$\frac{u_i - u_{i-1}}{\Delta x}$$

If $a < 0$

$$\frac{u_{i+1} - u_i}{\Delta x}$$

Explanations

- **Lie splitting** separates the 2D PDE into 1D steps
- First solve **advection in y**
- Then solve **advection in z**
- Uses **upwind finite difference**
- Derivative direction depends on flow direction
- Ensures numerical stability
- **First-order accurate in time**
- Computationally efficient implementation

STABILITY & IMPLEMENTATION

Equations

CFL Stability Condition

$$|a|\frac{\Delta t}{\Delta x} \leq 1$$

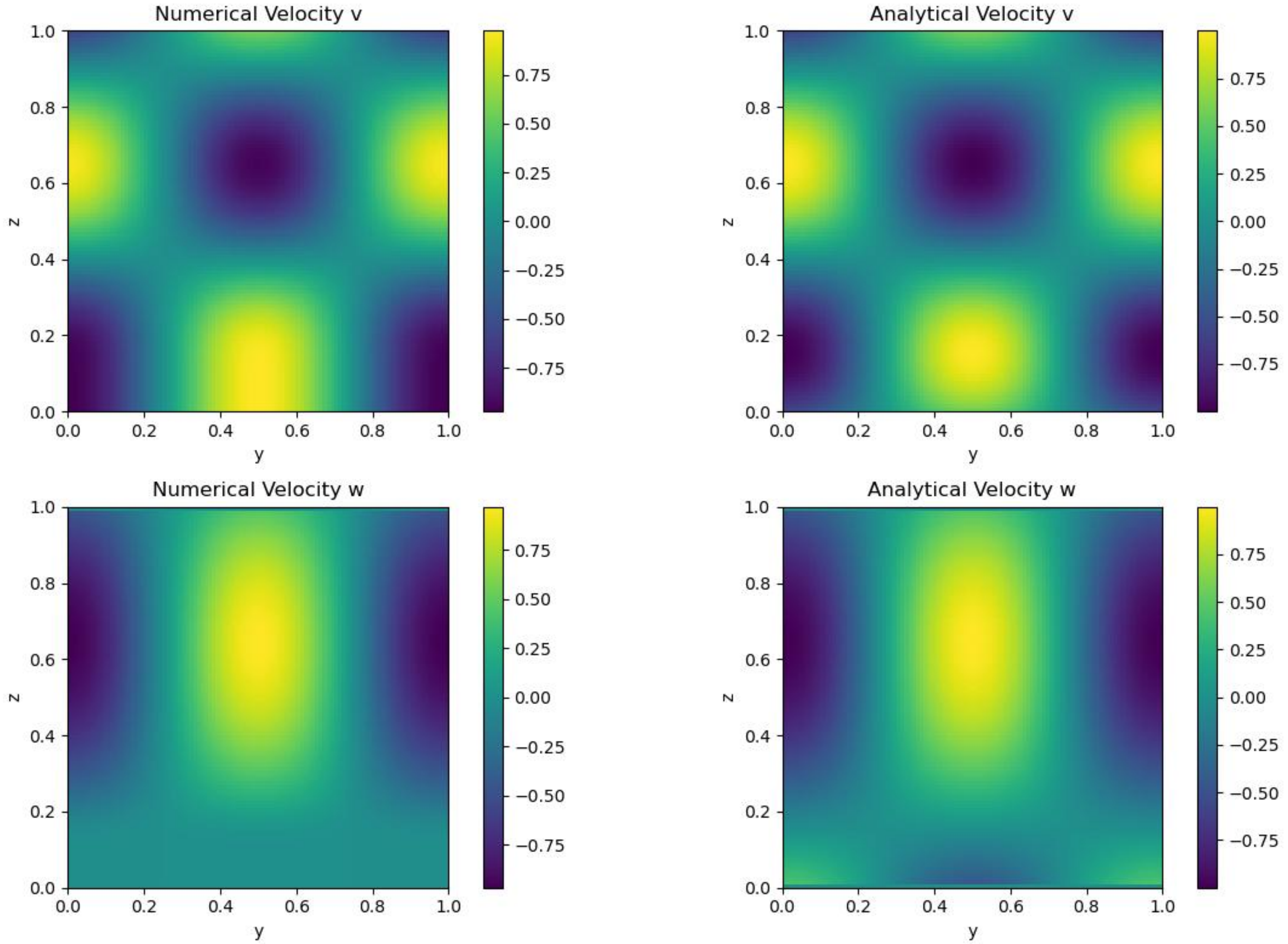
Time Step

$$\Delta t = 0.5 \min\left(\frac{\Delta y}{|a_y|}, \frac{\Delta z}{|a_z|}\right)$$

Explanations

- Stability controlled by **CFL condition**
- Time step chosen using **velocity magnitude**
- Grid resolution: **100 × 100 Cartesian grid**
- **Periodic boundary** in y implemented using `np.roll`
- **Wall boundary** in z direction
- Test case: constant velocity
a = (0.5 , 0.3)
- Initial condition: **sine / cosine wave**
- Numerical solution compared with **analytical solution**
- Observed **numerical diffusion**
- Maximum error ≈ **0.4**
- **First-order scheme ⇒ error decreases with grid refinement**

SIMULATION RESULTS



Incompressible Flow

PROBLEM SETUP

Equations

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \frac{1}{\rho}\nabla p = \nu\, \Delta \mathbf{u},$$
$$\nabla \cdot \mathbf{u} = 0,$$

$$\Omega = [0, L_y) \times [0, L_z]$$

$$\mathbf{u}(t,0,z) = \mathbf{u}(t,L_y,z) \qquad \forall\, t \geq 0,\ \ z \in [0, L_z],$$
$$w(t,y,0) = w(t,y,L_z) = 0 \qquad \forall\, t \geq 0,\ \ y \in [0, L_y).$$

$$\left.\frac{\partial v}{\partial z}\right|_{z=0,\, L_z} = 0,$$

Explanations

- velocity field : **u = (v,w)**, pressure : **p**
- **ρ** : fluid density, **ν** : the kinematic viscosity
- buoyancy parameter **β** is set to zero
- computational domain : two-dimensional rectangle **$\Omega = [0,L_y) \times [0,L_z]$**.
- **y** direction is periodic and the **z** direction is wall-bounded, reflecting a channel-like geometry.
- No fluid may penetrate the boundary **(no-penetration)**
- the tangential component v is subject to a **free-slip condition**

CHORIN PROJECTION METHOD

Equations

Incompressible Navier–Stokes

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \frac{1}{\rho}\nabla p = \nu \Delta \mathbf{u}$$

$$\nabla \cdot \mathbf{u} = 0$$

Predictor step

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left[-(\mathbf{u}^n \cdot \nabla)\mathbf{u}^n + \nu \Delta \mathbf{u}^n \right]$$

Explanations

- Used to solve **incompressible Navier–Stokes equations**
- Main idea: **decouple velocity and pressure**
- Computation split into three stages:
  1. **Predictor** → compute intermediate velocity **u***
  2. **Pressure solve** → enforce incompressibility
  3. **Projection** → correct velocity field
- Predictor step ignores **pressure gradient**
- Resulting velocity **u*** is generally **not divergence-free**

PRESSURE SOLVE & PROJECTION

Equations

Pressure Poisson equation

$$\Delta p = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*$$

Velocity correction

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p$$

Divergence constraint

$$\nabla \cdot \mathbf{u}^{n+1} = 0$$

Explanations

- Pressure obtained from **Poisson equation**
- RHS computed from **divergence of intermediate velocity**
- Poisson equation solved using **spectral method**
- **FFT → periodic direction**
- **DCT → wall-bounded direction**
- Projection step removes **irrotational component**
- Final velocity field becomes **divergence-free**

INITIAL CONDITIONS

Equations

$$\nabla \cdot \mathbf{u}_0 = 0$$

$$v_0(y, z) = \sin(n_y k_y y) \cos(n_z k_z z)$$

$$w_0(y, z) = -\frac{n_y k_y}{n_z k_z} \cos(n_y k_y y) \sin(n_z k_z z)$$

$$k_y = \frac{2\pi}{L_y}, \qquad k_z = \frac{\pi}{L_z}$$

$$\partial_y v_0 + \partial_z w_0 = 0$$

Implemented double mode

$$v_0 = \sin(k_y y) \cos(k_z z) + \sin(2k_y y) \cos(k_z z)$$

$$w_0 = -\frac{k_y}{k_z} \cos(k_y y) \sin(k_z z) - \frac{2k_y}{k_z} \cos(2k_y y) \sin(k_z z)$$

Explanations

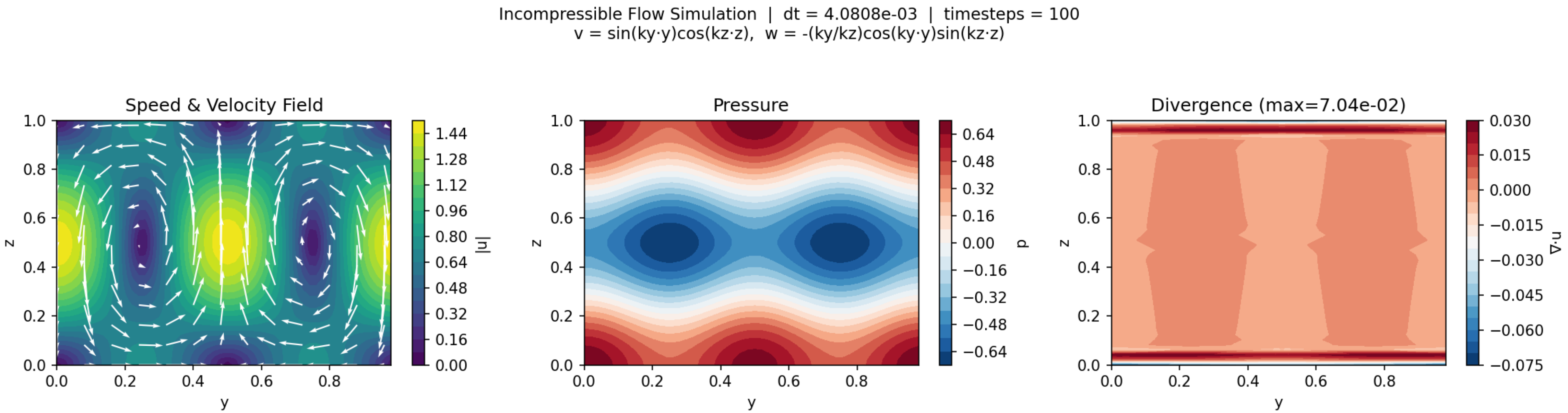
- Initial velocity field $\mathbf{u_0 = (v_0, w_0)}$
- Field is analytically **divergence-free**

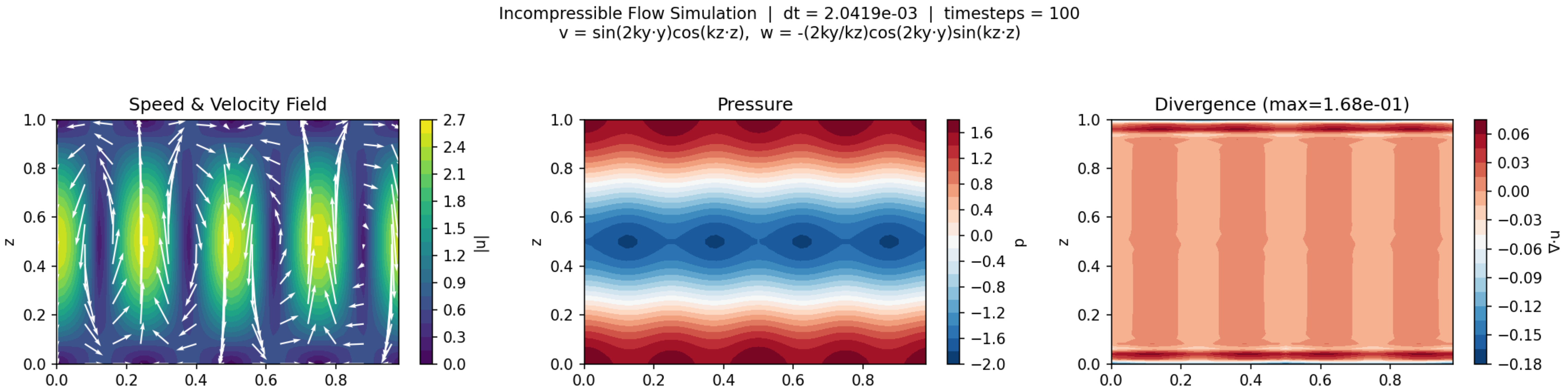
- ky ,kz are **fundamental wave numbers**
- ny nz determine **mode numbers**

- Ensures $\nabla \cdot \mathbf{u_0} = \mathbf{0}$ everywhere in the domain
- w0(y,0) = w0(y,Lz) = 0 → **no-penetration wall condition**
- y direction: periodic

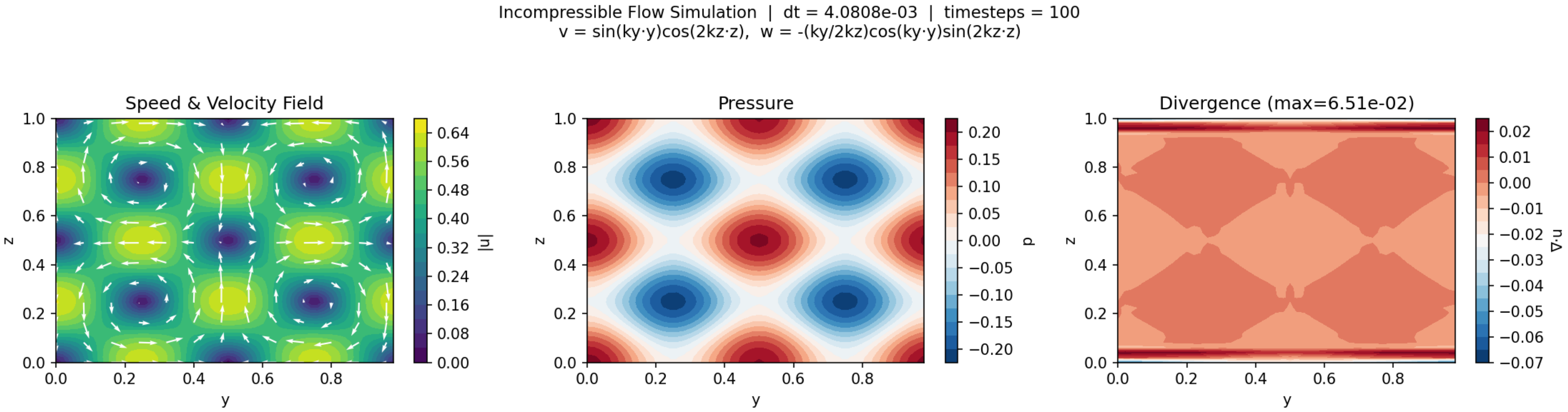
- Initial field built using **sinusoidal modes**
- A **double-mode superposition** is used in simulations

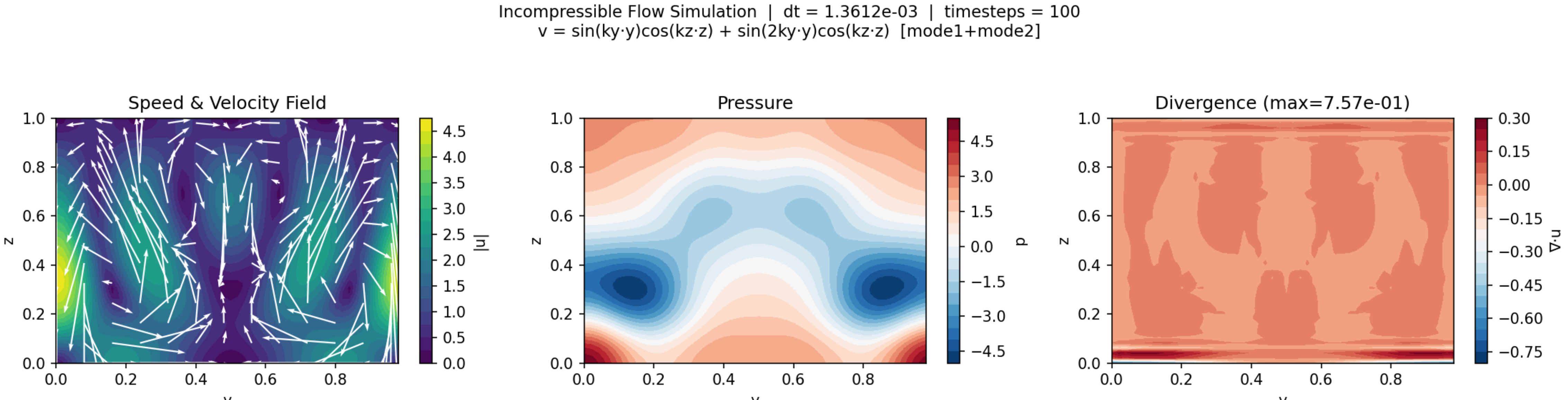
SIMULATION RESULTS





SIMULATION RESULTS





Rayleigh-Benard Convection

Equations

$$\partial_t T + \vec{u} \cdot \nabla T = \Delta T$$

$$\partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \nu \Delta \vec{u} + T \beta e_z$$

$$\nabla \cdot \vec{u} = 0$$

Numerical Method

- We've already implemented velocity solver with **Chorin's projection method.**

- Temperature solver:

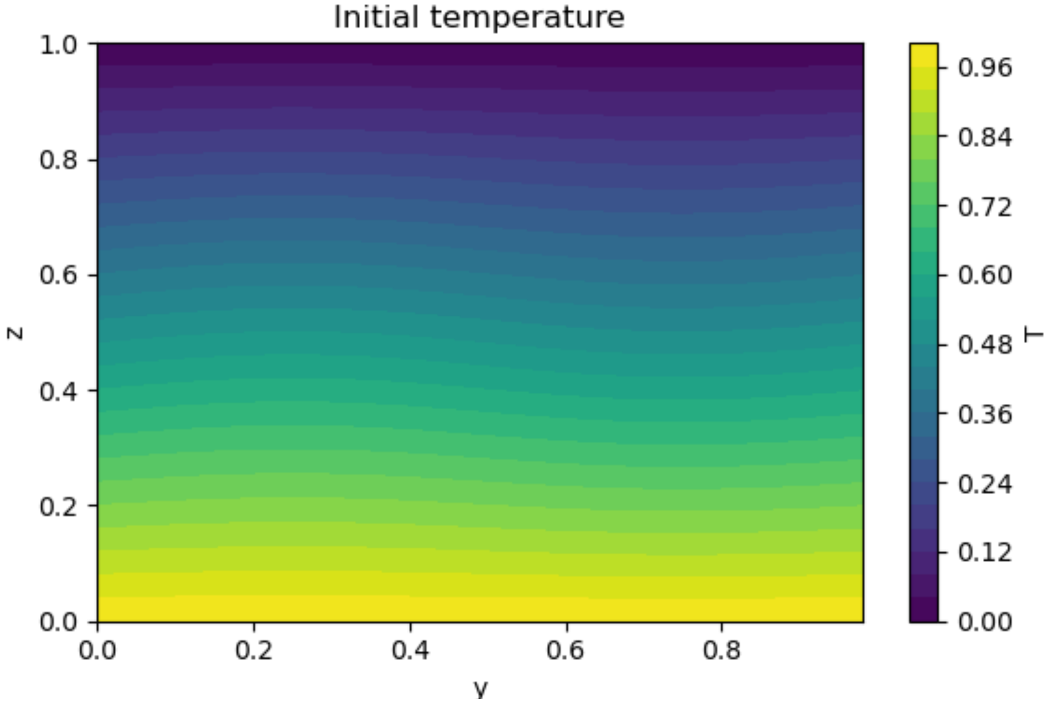
$$\partial_t T + \vec{u} \cdot \nabla T = \Delta T$$

Methods used:

- **Upwind scheme** for advection
- **Finite differences** for Laplacian
- **Explicit time stepping**

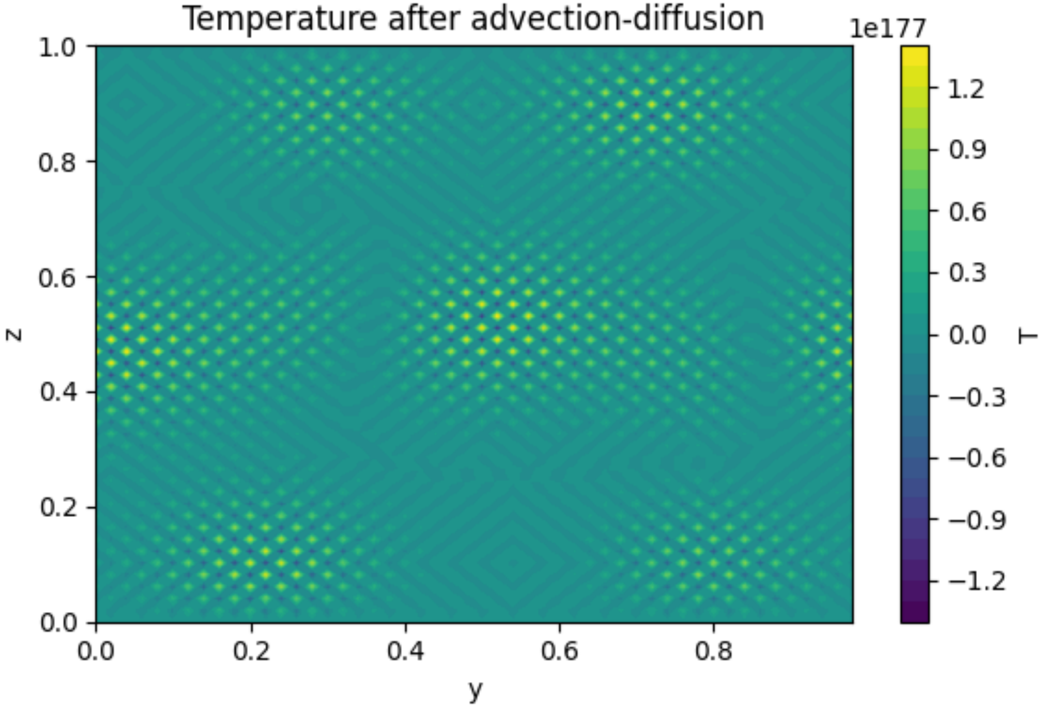
Initial Conditions

- domain $y \in [0, 1], z \in [0, 1]$
- periodic in horizontal direction y
- walls at top and bottom
- temperature $T = 1 - z + \epsilon \cos(2\pi y) \sin(\pi z)$



First Simulation Diagnostics

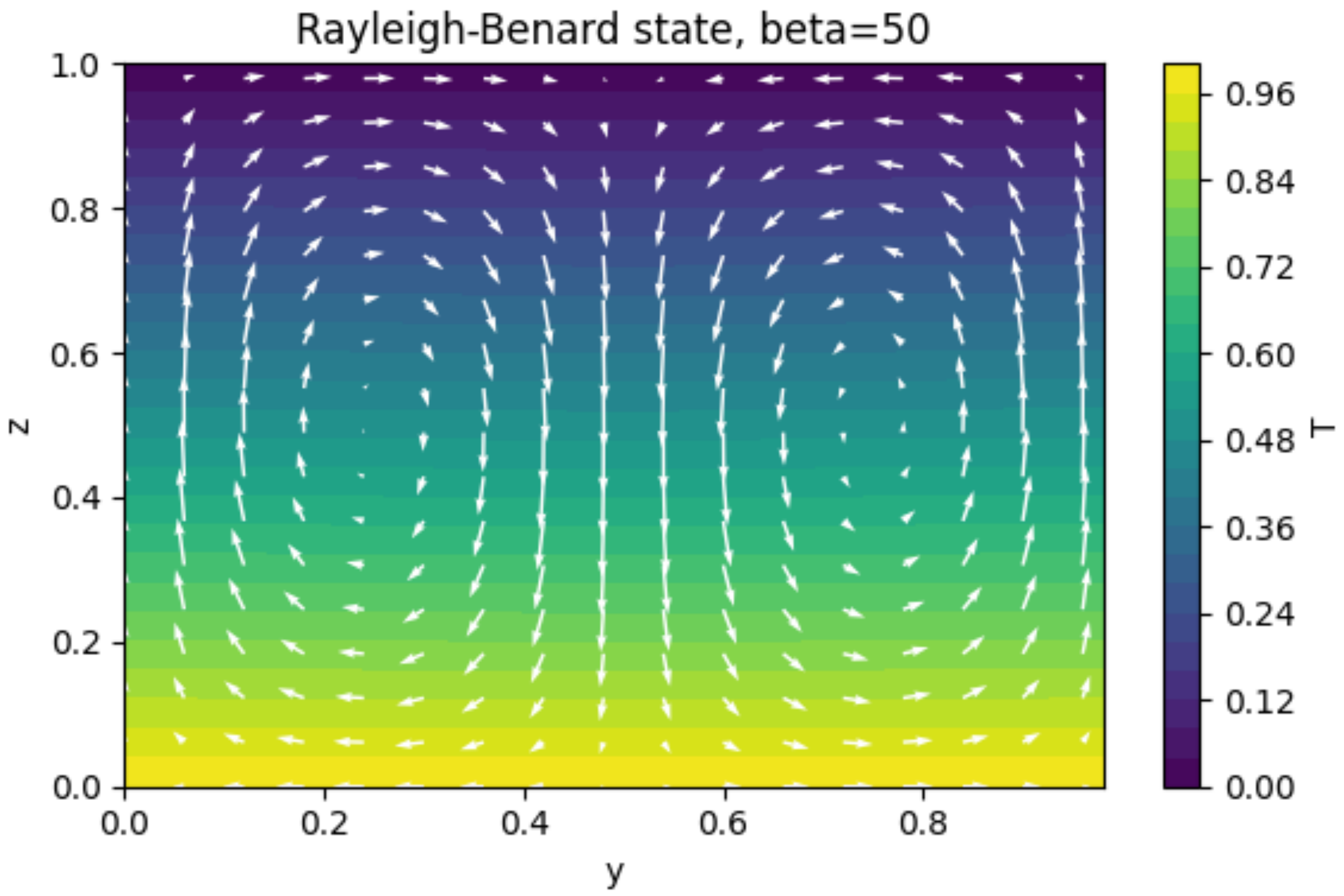
- fluid initially in motion (Chorin velocity field)
- solution diverges
- instable

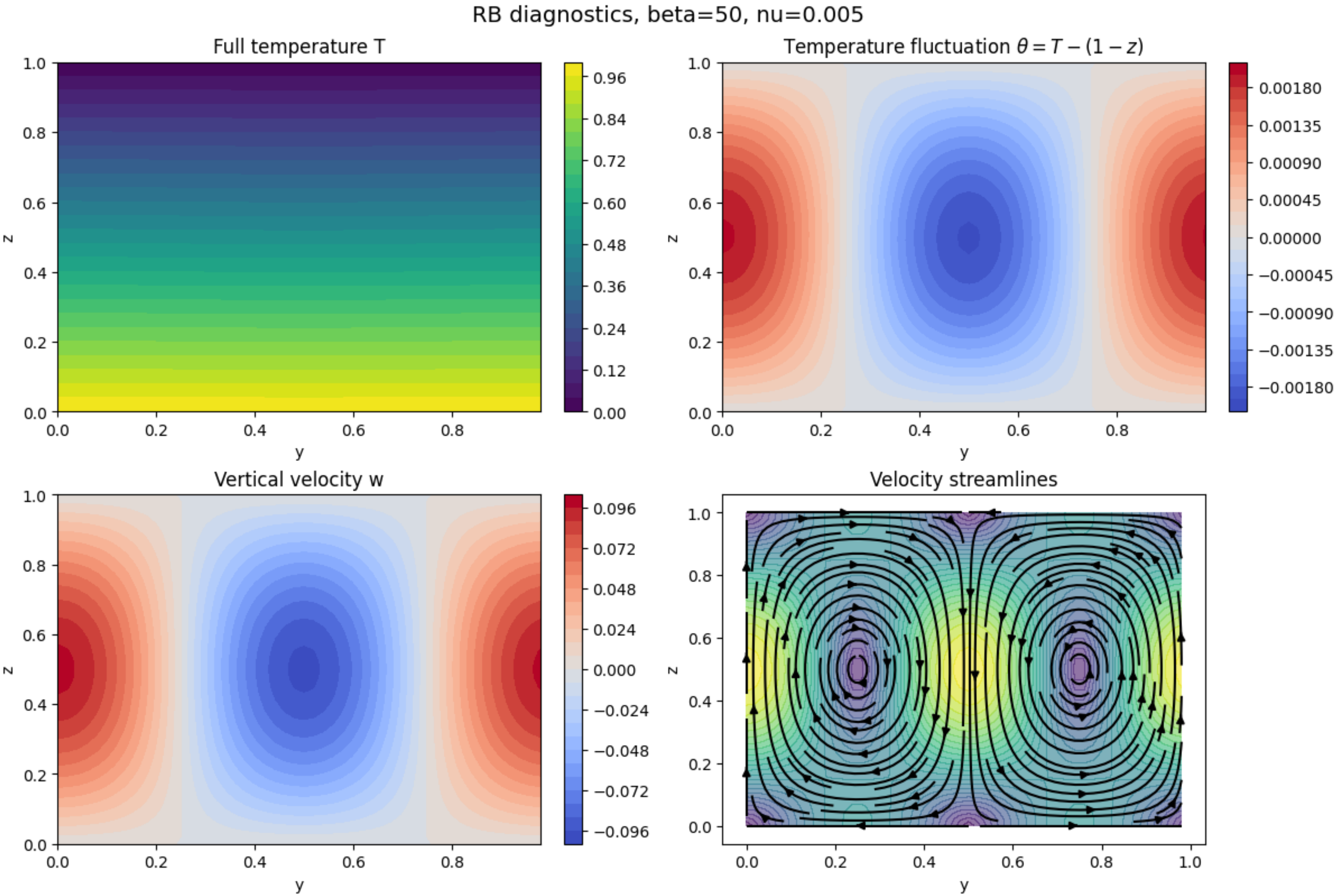


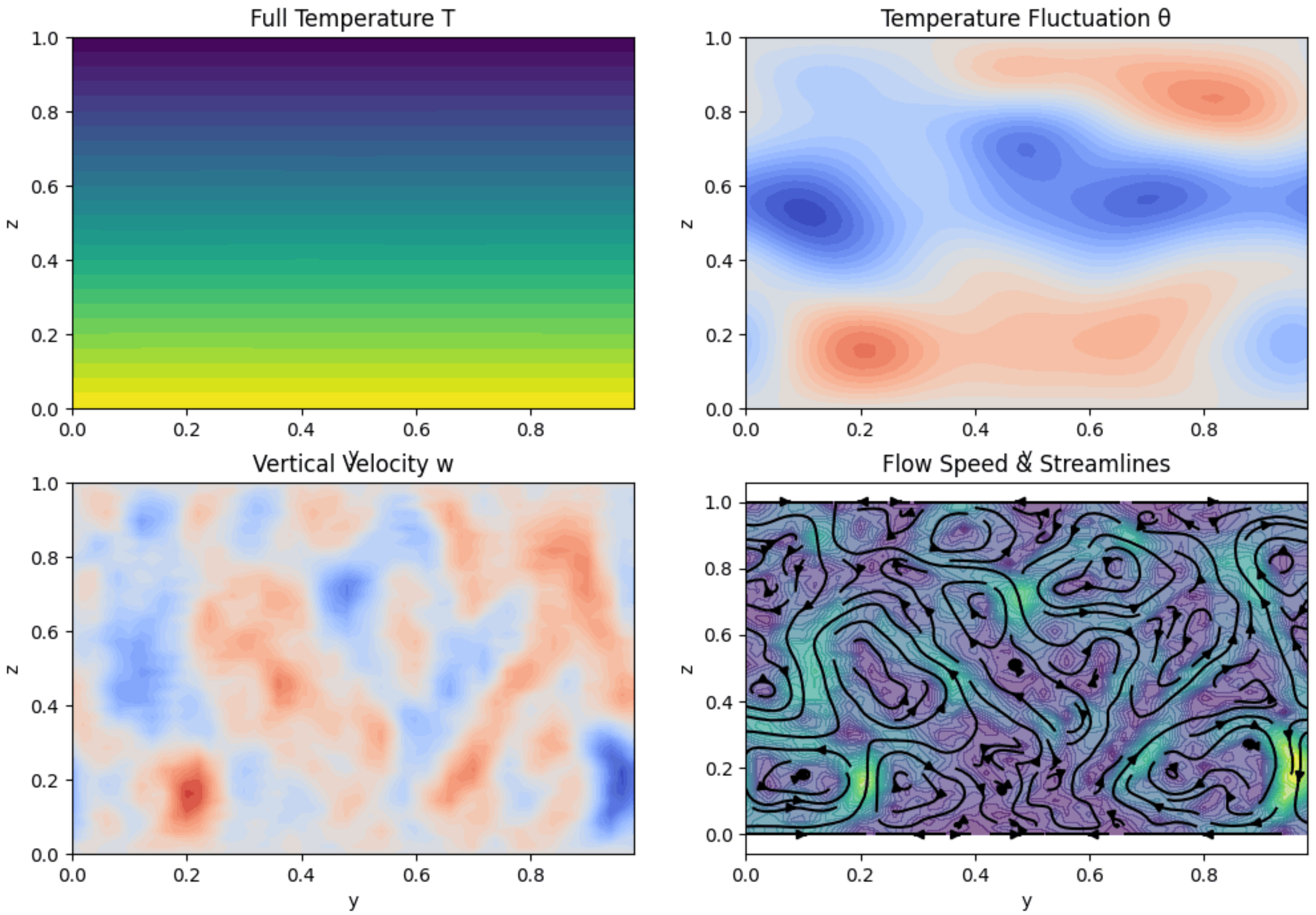
Solution: Velocity field initially zero

Observations:

- kinetic energy growth
- velocity magnitude over time
- T stable
- bouyancy drives motion




RB diagnostics, beta=50, nu=0.005
Full temperature T
Temperature fluctuation $\theta = T - (1 - z)$
Vertical velocity w
Velocity streamlines
y
z



Full Temperature T
Temperature Fluctuation θ
Vertical Velocity w
Flow Speed & Streamlines
z
y
0.0
0.2
0.4
0.6
0.8
1.0